

Perceptron

Introduction

Agha Ali Raza



Sources

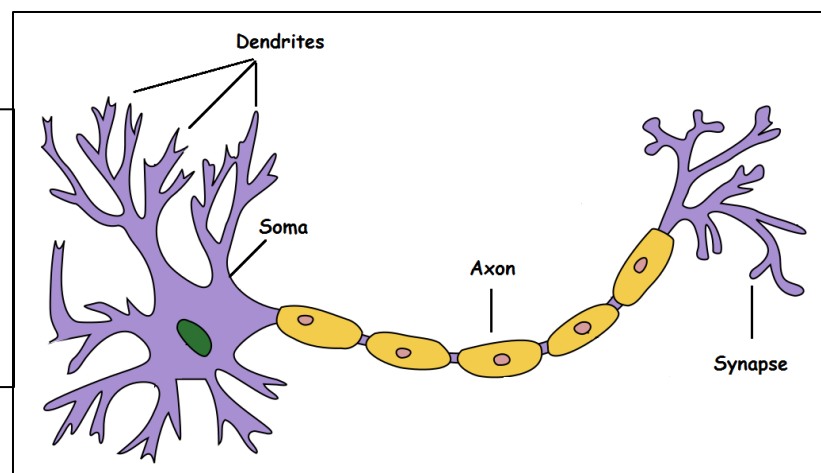
- **Akshay L Chandra – medium.com**
 - McCulloch-Pitts Neuron — Mankind's First Mathematical Model Of A Biological Neuron: <https://towardsdatascience.com/mcculloch-pitts-model-5fdf65ac5dd1>
 - Perceptron: The Artificial Neuron (An Essential Upgrade To The McCulloch-Pitts Neuron): <https://towardsdatascience.com/perceptron-the-artificial-neuron-4d8c70d5cc8d>
 - Perceptron Learning Algorithm: A Graphical Explanation Of Why It Works: <https://towardsdatascience.com/perceptron-learning-algorithm-d5db0deab975>
- **Prof. Mitesh M. Khapra** (<https://www.cse.iitm.ac.in/~miteshk/>) on NPTEL's (<http://nptel.ac.in/>) Deep Learning course (https://onlinecourses.nptel.ac.in/noc18_cs41/preview)
- **Machine Learning for Intelligent Systems**, Kilian Weinberger, Cornell, Lectures 3-6, https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecture_note03.html
- **Perceptrons. An Introduction to Computational Geometry.** Marvin Minsky and Seymour Papert. M.I.T. Press, Cambridge, Mass., 1969. <https://science.sciencemag.org/content/165/3895/780>

McCulloch-Pitts Neuron

(Akshay L Chandra: <https://towardsdatascience.com/mcculloch-pitts-model-5fdf65ac5dd1>)

- The fundamental unit of ANNs – an artificial neuron
- 1943 by McCulloch and Pitts: Mimicking the functionality of a biological neuron

Dendrites receive signals from other neurons
Soma processes the information
Axons transmit the output
Synapses are the connections to other neurons

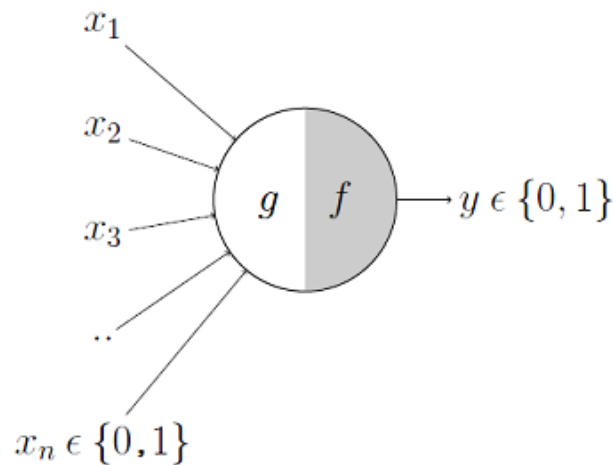


- About 86 billion of these in our brains on average!
- Each neuron gets activated/fired when its firing criteria is met
 - Based on the aggregation of signals from the inputs

McCulloch-Pitts Neuron

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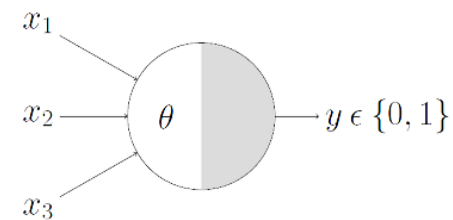
The first computational model of a neuron was proposed by Warren McCulloch (neuroscientist) and Walter Pitts (logician) in 1943.



$$g(x_1, x_2, x_3, \dots, x_n) = g(\mathbf{x}) = \sum_{i=1}^n x_i$$

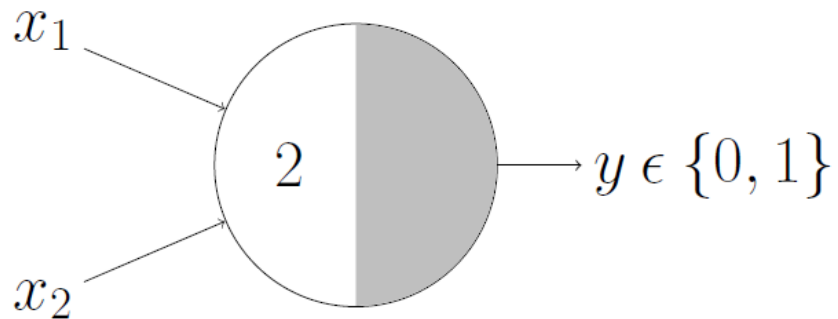
$$y = f(g(\mathbf{x})) = \begin{cases} 1 & \text{if } g(\mathbf{x}) \geq \theta \\ 0 & \text{if } g(\mathbf{x}) < \theta \end{cases}$$

- g aggregates inputs, f makes decisions
- θ is a hand-coded threshold
- All x_i are binary inputs and y_i is a binary output



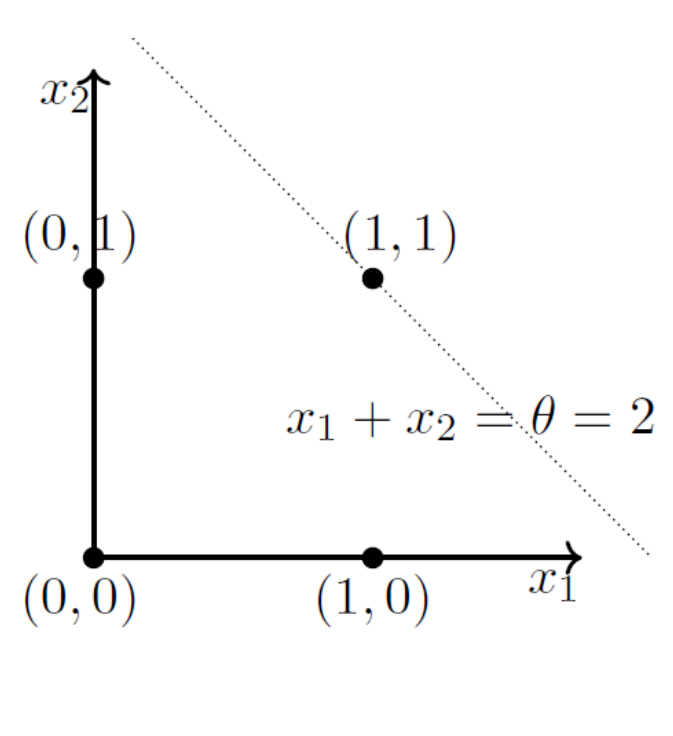
Boolean Functions

(Akshay L Chandra: <https://towardsdatascience.com/mcculloch-pitts-model-5fdf65ac5dd1>)



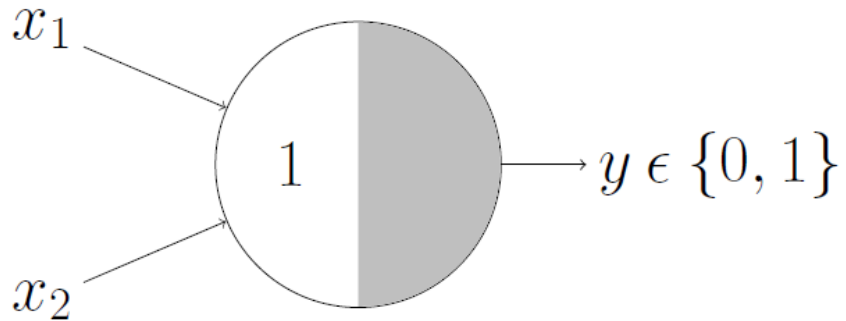
AND function

$$x_1 + x_2 = \sum_{i=1}^2 x_i \geq 2$$



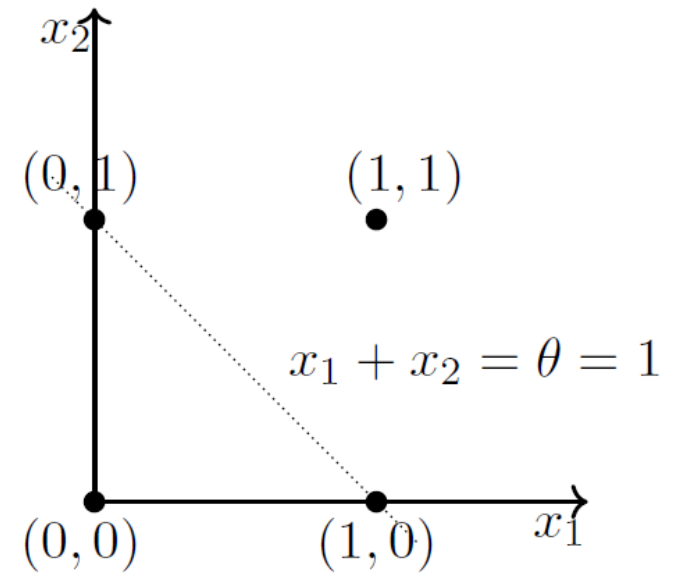
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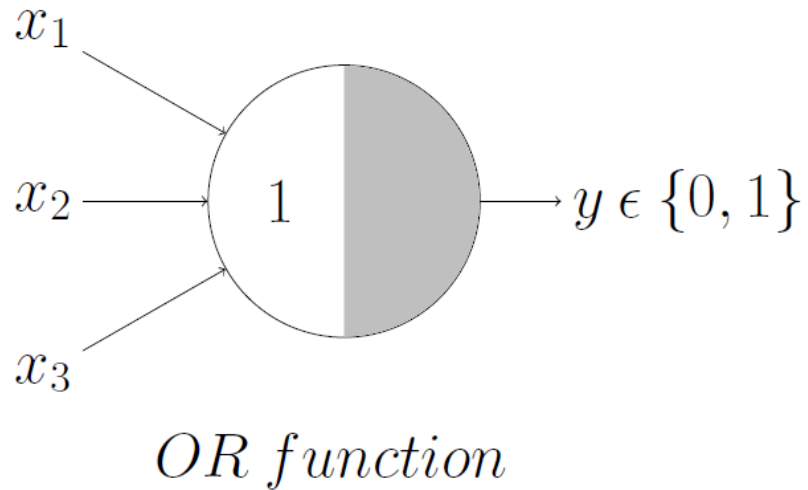
OR function

$$x_1 + x_2 = \sum_{i=1}^2 x_i \geq 1$$



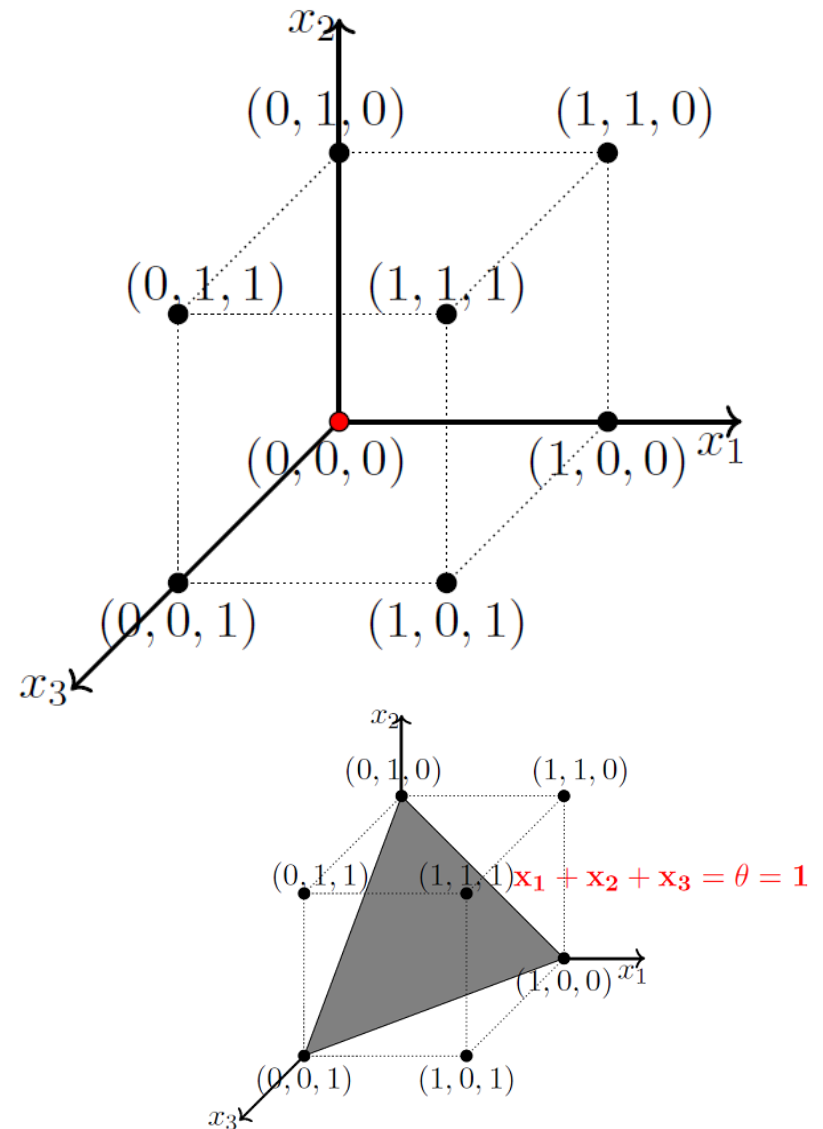
Boolean Functions

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$$x_1 + x_2 + x_3 = \sum_{i=1}^3 x_i \geq 1$$

The decision boundary would be a plane!



Limitations Of M-P Neuron

(Akshay L Chandra: <https://towardsdatascience.com/mcculloch-pitts-model-5fdf65ac5dd1>)

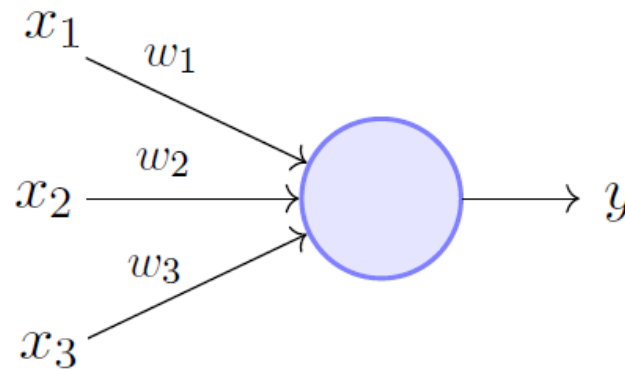
- Limited to Boolean inputs
- Hand-coded thresholds
- All inputs are equally important
 - Are all inputs born equal? Is number of legs as important of a input in a human vs. cat classifier as number of ears?
- What about functions that are not linearly separable? E.g. the XOR function.

In 1958, Frank Rosenblatt, an American psychologist, proposed the perception model

- Added weights and thresholds that could be learned. Allowed real-number inputs.

The Perceptron

(Akshay L Chandra: <https://towardsdatascience.com/perceptron-the-artificial-neuron-4d8c70d5cc8d>)

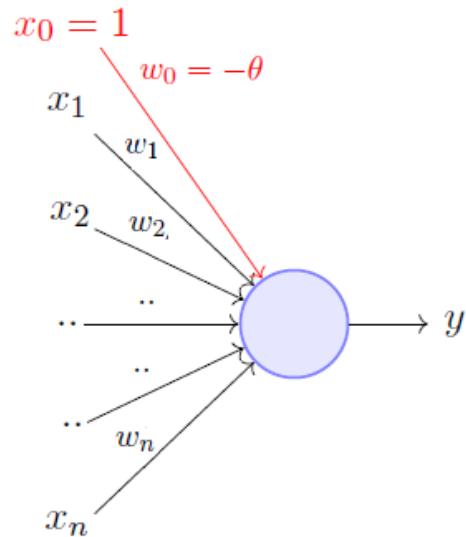
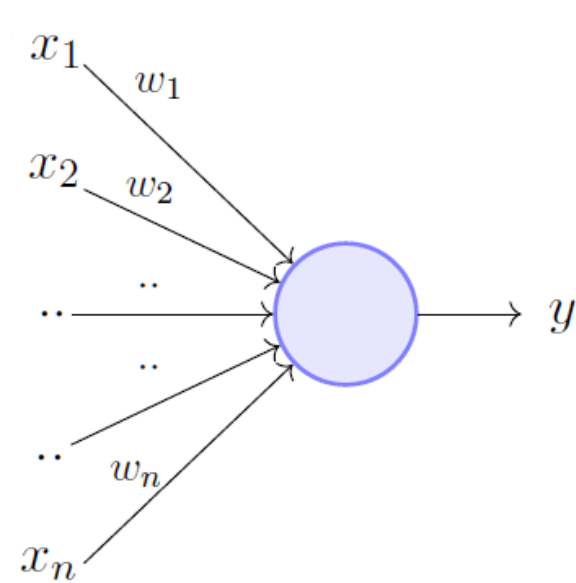


Perceptron Model (Minsky-Papert in 1969)

- Numerical weights associated with inputs
- No longer limited to Boolean inputs
- A learning mechanisms to train the weights and threshold

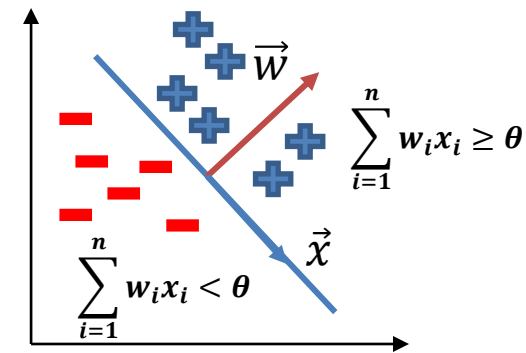
The Perceptron

(Akshay L Chandra: <https://towardsdatascience.com/perceptron-the-artificial-neuron-4d8c70d5cc8d>)



$$y = 1 \quad \text{if } \sum_{i=1}^n w_i * x_i \geq \theta$$

$$= 0 \quad \text{if } \sum_{i=1}^n w_i * x_i < \theta$$



What happens when $\sum_{i=1}^n w_i x_i = \theta$?

A more accepted convention,

$$y = 1 \quad \text{if } \sum_{i=0}^n w_i * x_i \geq 0$$

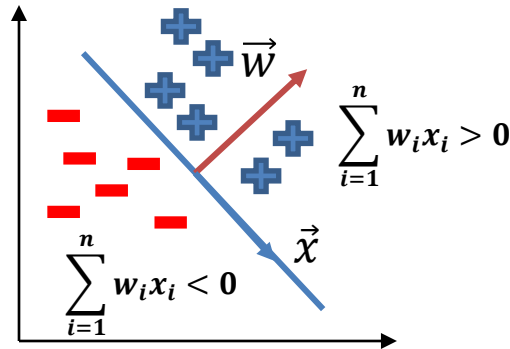
$$= 0 \quad \text{if } \sum_{i=0}^n w_i * x_i < 0$$

where, $x_0 = 1$ and $w_0 = -\theta$

The Perceptron

(Akshay L Chandra: <https://towardsdatascience.com/perceptron-the-artificial-neuron-4d8c70d5cc8d>)

- If we set $\theta = 0$, then the decision is simply: $h(x_i) = \text{sgn}(\mathbf{w}^T \mathbf{x}_i + \theta)$
- After absorbing θ this becomes: $h(x_i) = \text{sgn}(\mathbf{w}^T \mathbf{x}_i)$



- This also simplifies defining correct/incorrect classification:
 $y_i(\mathbf{w}^T \mathbf{x}_i) > 0 \Leftrightarrow x_i \text{ is classified correctly}$
 $y_i > 0 \text{ and } (\mathbf{w}^T \mathbf{x}_i) > 0 \text{ OR } y_i < 0 \text{ and } (\mathbf{w}^T \mathbf{x}_i) < 0$

The OR function

(Akshay L Chandra: <https://towardsdatascience.com/perceptron-the-artificial-neuron-4d8c70d5cc8d>)

x_1	x_2	OR	
0	0	0	$w_0 + \sum_{i=1}^2 w_i x_i < 0$
1	0	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$
0	1	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$
1	1	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$

$$w_0 + w_1 \cdot 0 + w_2 \cdot 0 < 0 \implies w_0 < 0$$

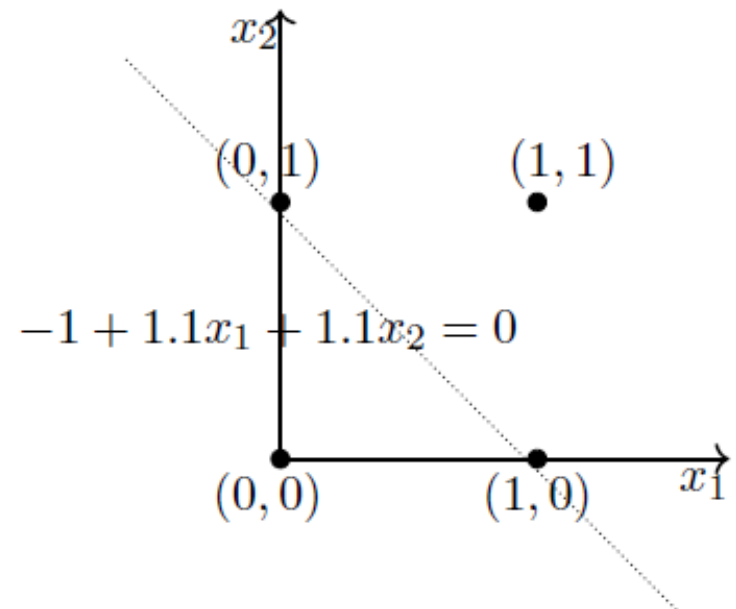
$$w_0 + w_1 \cdot 0 + w_2 \cdot 1 \geq 0 \implies w_2 > -w_0$$

$$w_0 + w_1 \cdot 1 + w_2 \cdot 0 \geq 0 \implies w_1 > -w_0$$

$$w_0 + w_1 \cdot 1 + w_2 \cdot 1 \geq 0 \implies w_1 + w_2 > -w_0$$

One possible solution is

$$w_0 = -1, w_1 = 1.1, w_2 = 1.1$$



The XOR function

(Akshay L Chandra: <https://towardsdatascience.com/perceptron-the-artificial-neuron-4d8c70d5cc8d>)

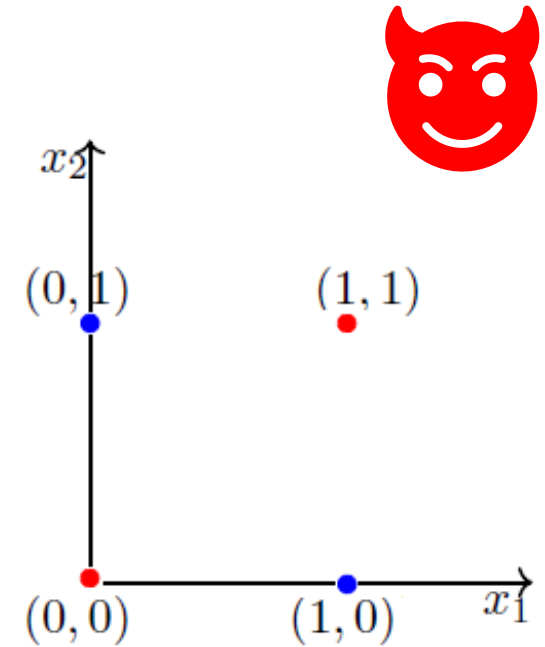
x_1	x_2	XOR	
0	0	0	$w_0 + \sum_{i=1}^2 w_i x_i < 0$
1	0	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$
0	1	1	$w_0 + \sum_{i=1}^2 w_i x_i \geq 0$
1	1	0	$w_0 + \sum_{i=1}^2 w_i x_i < 0$

$$w_0 + w_1 \cdot 0 + w_2 \cdot 0 < 0 \implies w_0 < 0$$

$$w_0 + w_1 \cdot 0 + w_2 \cdot 1 \geq 0 \implies w_2 > -w_0$$

$$w_0 + w_1 \cdot 1 + w_2 \cdot 0 \geq 0 \implies w_1 > -w_0$$

$$w_0 + w_1 \cdot 1 + w_2 \cdot 1 \geq 0 \implies w_1 + w_2 < -w_0$$

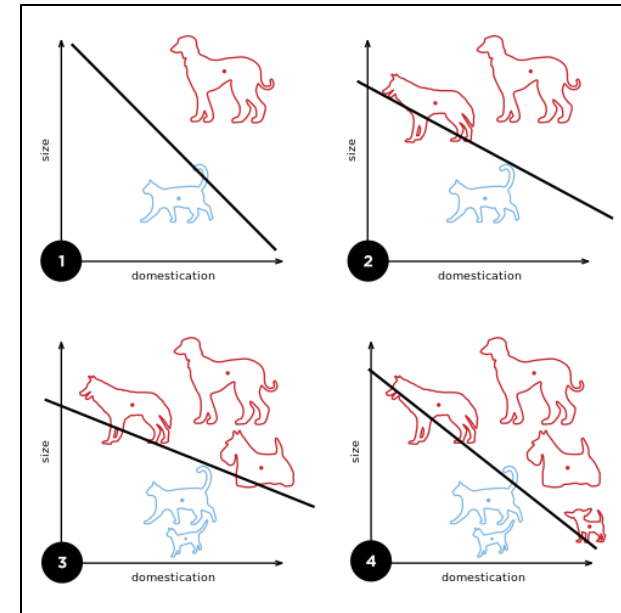


- The fourth equation contradicts the second and the third equation
- There are no perceptron solutions for non-linearly separated data
- A single perceptron cannot learn to separate the data that are non-linear in nature
- In the book published by Minsky and Papert in 1969, the authors implied that, since a single artificial neuron is incapable of implementing some functions such as the XOR logical function, larger networks also have similar limitations, and therefore should be dropped.
- Later research on three-layered perceptions showed how to implement such functions, therefore saving the technique from obliteration.

The Perceptron

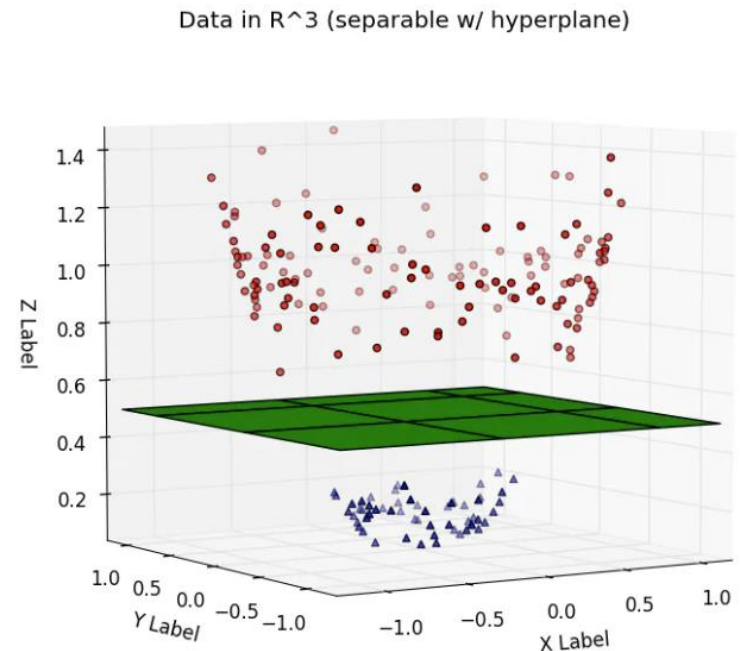
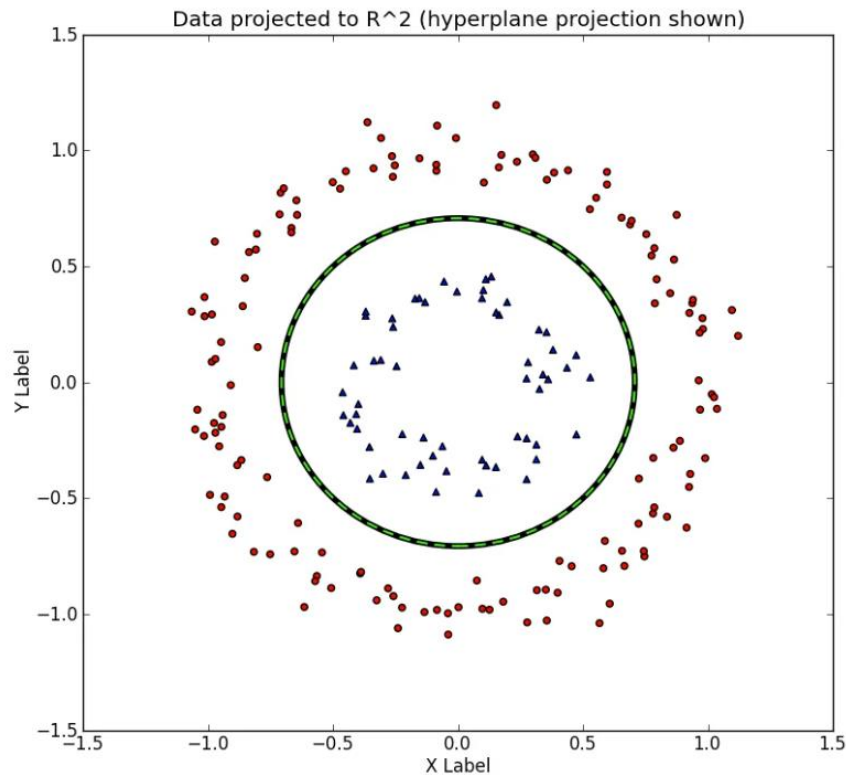
(Akshay L Chandra: <https://towardsdatascience.com/perceptron-the-artificial-neuron-4d8c70d5cc8d>)

- A perceptron separates the input space into two halves, positive and negative.
- All the inputs that produce *true* lie on one side (positive half space) and all the inputs that produce *false* lie on the other side (negative half space)
- A single perceptron can only be used to implement linearly separable functions
 - Just like the M-P neuron
- The differences:
 - The inputs can be assigned different importance
 - The weights and the threshold can be learned
 - The inputs can be real values



Adding dimensions to achieve linear separability

(The Kernel Trick in Support Vector Classification, Drew Wilimitis,
<https://towardsdatascience.com/the-kernel-trick-c98cdbcaeb3f>)

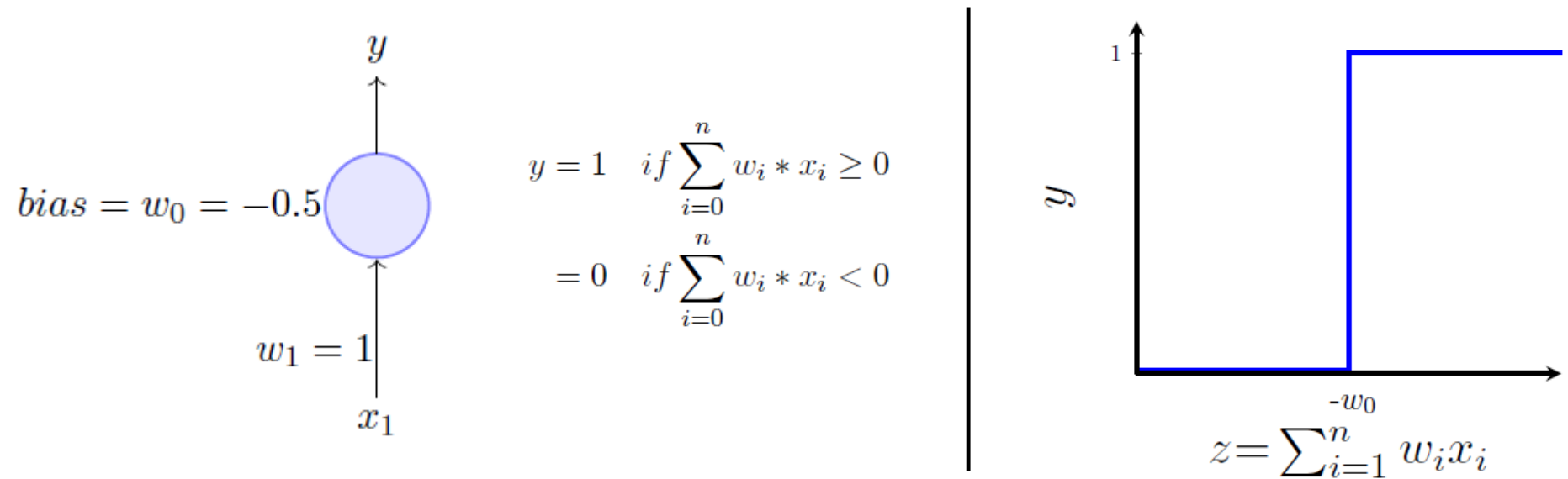


Adding a third dimension: Each point's distance from the center

Connecting the dots

(Akshay L Chandra: <https://towardsdatascience.com/perceptron-the-artificial-neuron-4d8c70d5cc8d>)

- The perceptron employs very harsh thresholds!
- E.g. if the threshold is 0.5: Input = 0.49 would yield a negative and input=0.51 would yield a positive output

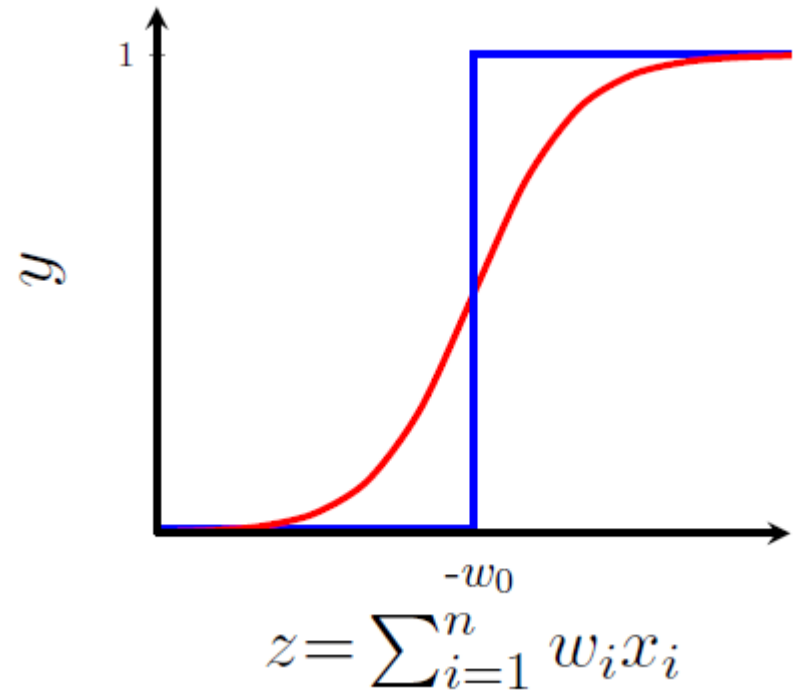


Connecting the dots – Other Activation Functions

(Akshay L Chandra: <https://towardsdatascience.com/perceptron-the-artificial-neuron-4d8c70d5cc8d>)

- Hence, the need for smoother functions like the sigmoid!

$$y = \frac{1}{1 + e^{-(w_0 + \sum_{i=1}^n w_i x_i)}}$$



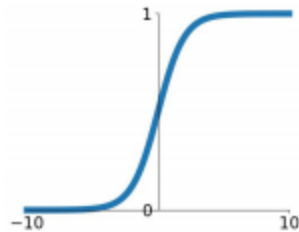
- No more sharp transition around the w_0
- The output is no longer binary but a real value between 0 and 1 which can be interpreted as a probability
- So instead of yes/no decision, we get the probability of yes.
- The output is **smooth**, **continuous** and **differentiable**.

Other Activation Functions (“squishifiers”)

(Stanford: http://cs231n.stanford.edu/slides/2017/cs231n_2017_lecture6.pdf)

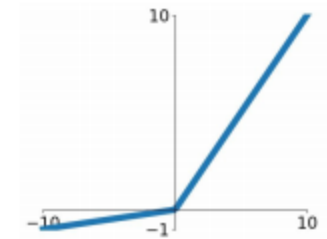
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



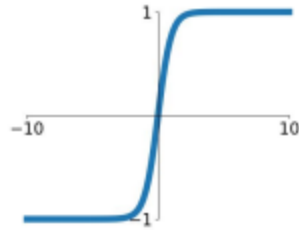
Leaky ReLU

$$\max(0.1x, x)$$



tanh

$$\tanh(x)$$

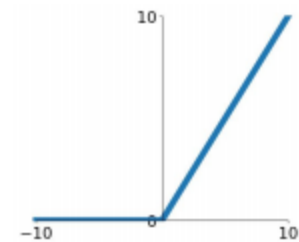


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

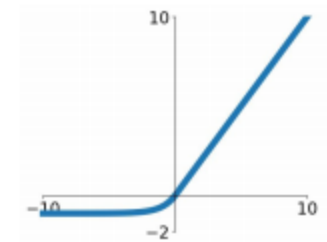
ReLU

$$\max(0, x)$$



ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



Up next

- Perceptron Learning Algorithm
 - Geometric Intuition of why the learning works

Perceptron

Learning Algorithm

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Perceptron Algorithm

(Machine Learning for Intelligent Systems, Kilian Weinberger, Cornell, Lectures 3-6,
<https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote03.html>)

Initialize $\vec{w} = 0$

while true do

$m = 0$

for $(x_i, y_i) \in D$ **do**

if $y_i(\vec{w}^T \cdot \vec{x}_i) \leq 0$ **then**

$\vec{w} \leftarrow \vec{w} + y_i \vec{x}_i$

$m \leftarrow m + 1$

end if

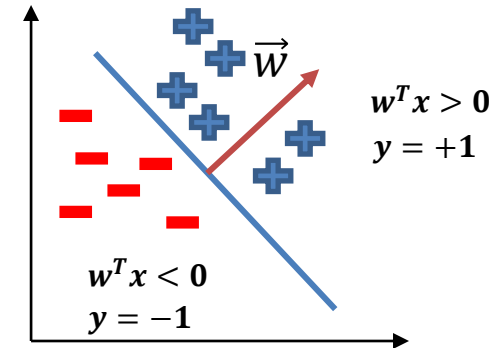
end for

if $m = 0$ **then**

break

end if

end while



Correct classification

$y_i = +1, w^T x_i > 0$

$y_i = -1, w^T x_i < 0$

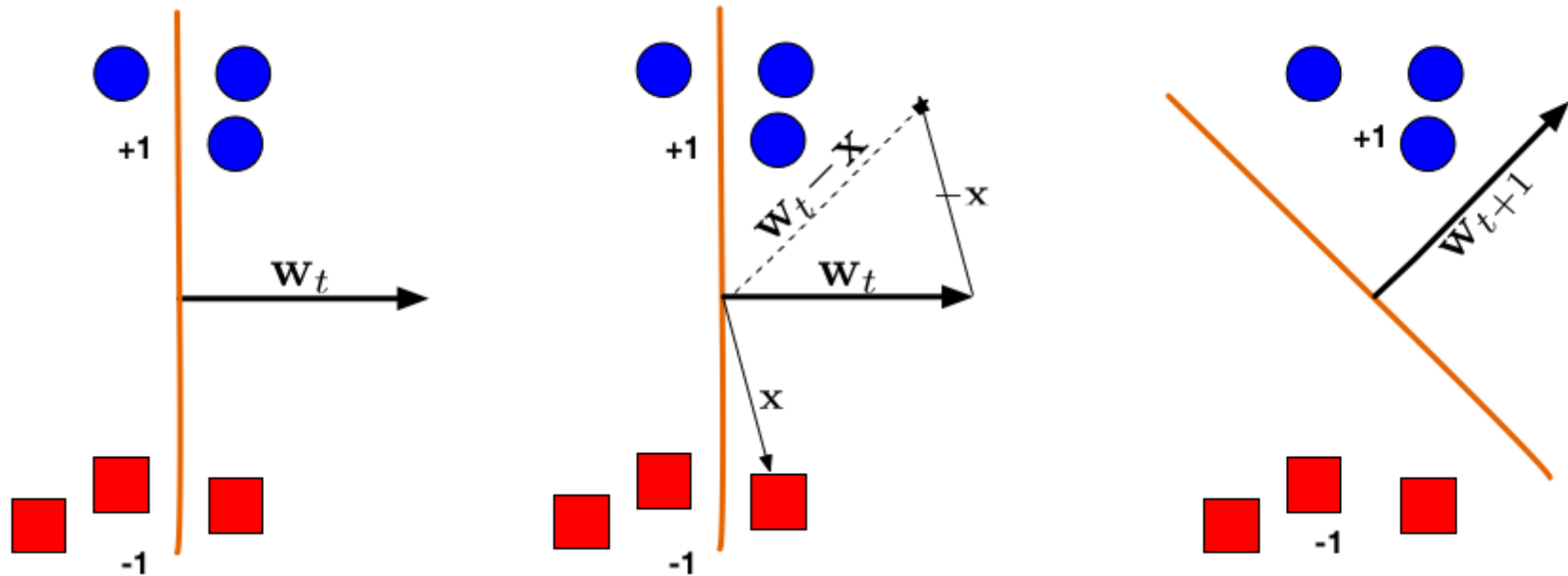
Incorrect classification

$y_i = +1, w^T x_i < 0$

$y_i = -1, w^T x_i > 0$

Geometric Interpretation

(Machine Learning for Intelligent Systems, Kilian Weinberger, Cornell, Lectures 3-6,
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(Left:) The hyperplane defined by w_t misclassifies one red (-1) and one blue (+1) point.

(Middle:) The red point x is chosen and used for an update. Because its label is -1 we need to subtract x from w_t .

(Right:) The updated hyperplane $w_{t+1} = w_t - x$ separates the two classes and the Perceptron algorithm has converged.

Up next

- Perceptron Convergence
 - Do we have any guarantee that the algorithm will converge given the data is linearly separable?

Perceptron

Learning Algorithm – Proof of Convergence

Agha Ali Raza



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$\vec{w} \leftarrow \vec{w} + y_i \vec{x}_i$

$m \leftarrow m + 1$

end if

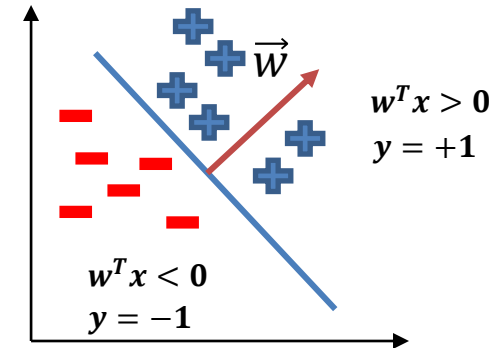
end for

if $m = 0$ **then**

break

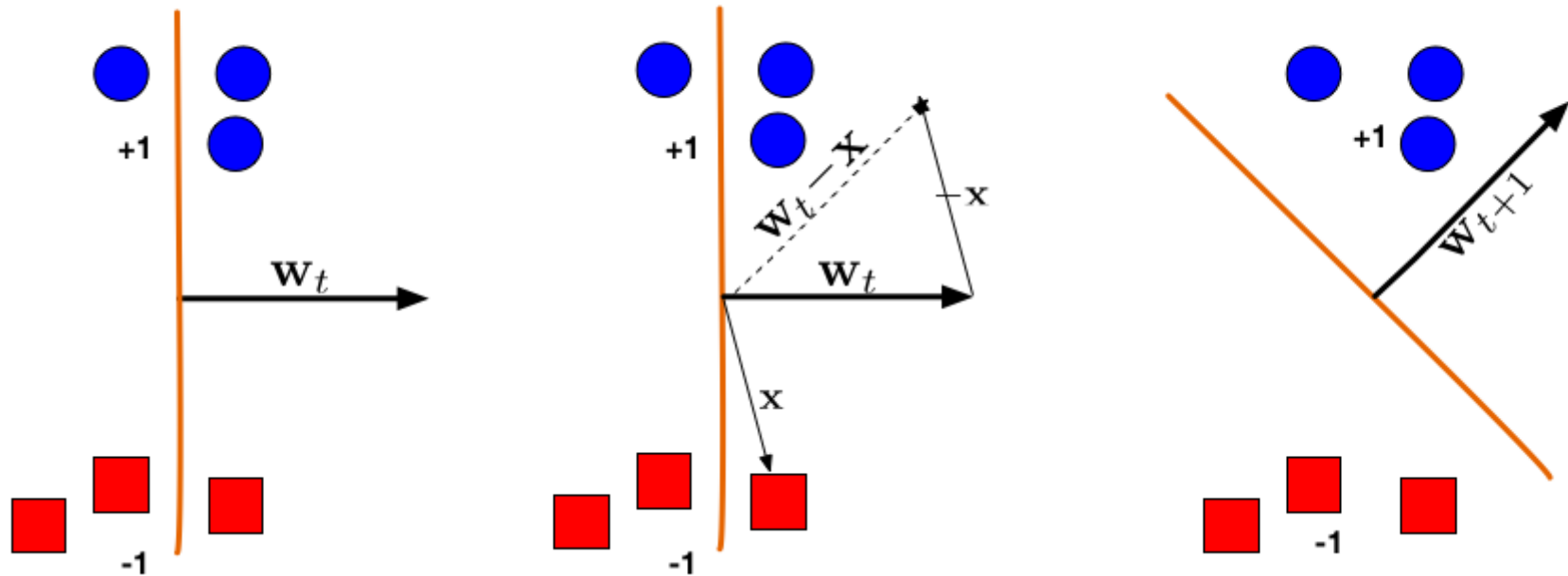
end if

end while



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Perceptron Learning Algorithm Convergence

(Machine Learning for Intelligent Systems, Kilian Weinberger, Cornell, Lectures 3-6,
<https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote03.html>)

- **Assumption:** The data is linearly separable
- **Goal:** If a data set is linearly separable, the Perceptron will find a separating hyperplane in a finite number of updates
 - If the data is not linearly separable, it will loop forever
- **Strategy:** Show that the algorithm monotonically makes *progress* towards a correct solution.
- **Correct solution:** A hyperplane w^* that separates the negative and positive examples:

$$\exists w^* \text{ s.t. } y_i(x_i^\top w^*) > 0 \quad \forall (x_i, y_i) \in D$$

Perceptron Learning Algorithm Convergence

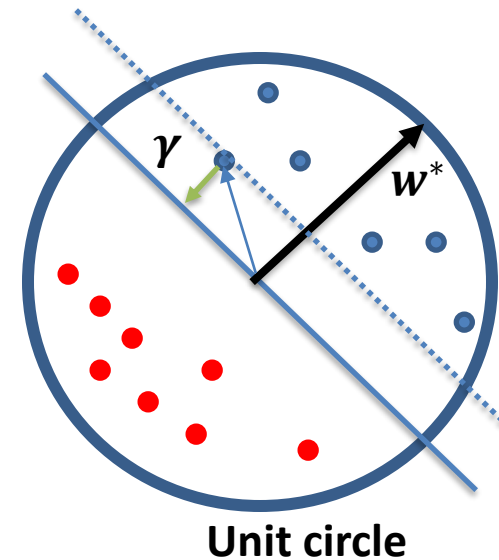
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Now, suppose that we rescale each data point and the w^* such that:

$$\|w^*\| = 1 \text{ and } \|x_i\| \leq 1 \forall x_i \in D$$

Let us define the Margin γ of the hyperplane w^* as:

$$\gamma = \min_{(x_i, y_i) \in D} |x_i^\top w^*|$$



To summarize our setup:

- All inputs x_i live within the unit sphere
- There exists a separating hyperplane defined by w^* , with $\|w^*\| = 1$
 - i.e. w^* lies exactly on the unit sphere
- γ is the distance from the hyperplane to the closest data point

Perceptron Learning Algorithm Convergence

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- **Theorem:** If all of the above holds, then the Perceptron algorithm makes at most $\frac{1}{\gamma^2}$ mistakes.

Proof:

- Consider the effect of an update (w becomes $w + yx$) on the two terms $\mathbf{w}^\top \mathbf{w}^*$ and $\mathbf{w}^\top \mathbf{w}$.
- We will use two facts:
 - $y(x^\top w) \leq 0$: This holds because x is misclassified by w - otherwise we wouldn't make the update.
 - $y(x^\top w^*) > 0$: This holds because w^* is a separating hyper-plane and classifies all points correctly.

Perceptron Learning Algorithm Convergence

(Machine Learning for Intelligent Systems, Kilian Weinberger, Cornell, Lectures 3-6,
<https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote03.html>)

Consider the effect of an update on $w^\top w^*$:

$$(w + yx)^\top w^* = w^\top w^* + y(x^\top w^*) \geq w^\top w^* + \gamma$$

- The inequality follows from the facts that
 - $y(x^\top w^*) > 0$: This holds because w^* is a separating hyper-plane and classifies all points correctly
 - For w^* , the distance from the hyperplane defined by w^* to x must be at least γ
 - i.e. $y(x^\top w^*) = |x^\top w^*| \geq \gamma$
- **This means that for each update, $w^\top w^*$ grows by at least γ**

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Consider the effect of an update on $w^\top w$:

$$(w + yx)^\top (w + yx) = w^\top w + yw^\top x + ywx^\top + y^2 x^\top x$$

$$w^\top w + 2y(w^\top x) + y^2(x^\top x) \leq w^\top w + 1$$

The inequality follows from the facts that:

- $2y(w^\top x) < 0$ as we had to make an update, meaning x was misclassified
- $0 \leq y^2(x^\top x) \leq 1$ as $y^2 = 1$ and all $x^\top x \leq 1$ (because this gives the magnitude of x and we know that $\|x\| \leq 1$)
- **This means that for each update, $w^\top w$ grows by at most 1**

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Now we know that after M updates the following two inequalities must hold:

$$(1) w^\top w^* \geq M\gamma$$

$$(2) w^\top w \leq M$$

We can then complete the proof:

$$\begin{aligned} M\gamma &\leq w^\top w^* \\ &= \|w\| \|w^*\| \cos(\theta) = \|w\| \cos(\theta) \\ &\leq \|w\| \\ &= \sqrt{w^\top w} \\ &\leq \sqrt{M} \end{aligned}$$

$$\begin{aligned} \Rightarrow M\gamma &\leq \sqrt{M} \\ \Rightarrow M^2\gamma^2 &\leq M \\ \Rightarrow M &\leq \frac{1}{\gamma^2} \end{aligned}$$

by (1)

as $\|w^\| = 1$*

as $\cos(\theta) \leq 1$

by the definition of $\|w\|$

by (2)

The number of updates M is bounded from above by a constant.

Now, can you answer the question about the time to convergence in case of various margin sizes?



For more details please visit
<http://aghaaliraza.com>



Thank you!